

# DUET-VO → RA-L

Turning an IROS rejection into a journal submission

|         |                                  |           |                                     |
|---------|----------------------------------|-----------|-------------------------------------|
| TARGET  | IEEE RA-L, rolling               | SOURCE    | IROS 2026 #651 (rejected, 36% year) |
| SCOPE   | Stereo event VO, no loop closure | COMPANION | Paper 2: backbone-native LC         |
| WRITTEN | 2026-08-05                       | ESTIMATE  | 6–8 weeks (team), 10–14 (solo)      |

## Milestone dashboard

| PHASE | MILESTONE                                   | PROGRESS                   | EFFORT | DONE                                |
|-------|---------------------------------------------|----------------------------|--------|-------------------------------------|
| P0    | Foundations — harness, literature DB, plan  | <div><div></div></div> 3/3 | done   | <input checked="" type="checkbox"/> |
| P1    | De-risk — builds, datasets, author requests | <div><div></div></div> 0/4 | 2–3 wk | <input type="checkbox"/>            |
| P2    | Accuracy evidence — the main tables         | <div><div></div></div> 0/5 | 2 wk   | <input type="checkbox"/>            |
| P3    | Runtime evidence — the centrepiece          | <div><div></div></div> 0/5 | 1–2 wk | <input type="checkbox"/>            |
| P4    | Ablations                                   | <div><div></div></div> 0/4 | 1 wk   | <input type="checkbox"/>            |
| P5    | Manuscript & submission                     | <div><div></div></div> 0/6 | 3–4 wk | <input type="checkbox"/>            |

### Single point of failure — resolve in week 1

**S-DEVO must build and run on the AGX Orin.** PyTorch + TensorRT + their ROS stack on aarch64 is genuinely hostile, and if that build fails the primary head-to-head result disappears — the entire runtime story rests on it.

Start it *before* downloading a single dataset. Everything else in this plan has a fallback; this does not. Fallback if it proves impossible: their published Orin NX figure plus our power-capped AGX Orin measurement as the bridge (M3.4).

☒ done ☐ open ☒ blocked / critical path

## 1 Why it was actually rejected

Not novelty. All three reviewers liked the idea; R7 explicitly praised the contribution framing and the methodology section. Two things sank it.

- Evidence provenance.** Table II is  $\approx 90$  cells:  $\approx 40$  copied from S-DEVO, 27 bare dashes, 23 self-run — and the 23 self-run cells are exactly where DUET-VO wins. R5 named the tell: “*even the worst results in the DUET-VO ablation study are better than the reported results of S-DEVO on M3ED.*” That reads as cherry-picking regardless of intent, and it is unanswerable without re-running.
- The efficiency claim had no head-to-head data.** The headline benefit is speed; Table IV has a single column of our own numbers on our own hardware, missing feature extraction and undistortion. The one comparative sentence in the paper contradicted a published number with no measurement behind it.

Everything else in the reviews is mechanical.

### The rule that follows from R5’s objection

Quoting published numbers is standard practice — S-DEVO does it too. What is fatal is quoting *selectively*, so that the provenance boundary coincides with the performance boundary. Borrowed where we look ordinary, self-run where we dominate. No reviewer can unsee that pattern, and no amount of good faith

changes how it reads.

## 2 The reframe

Our objections to the reviewers are scientifically correct but were **never entered as evidence**. “ATE on a 5 cm handheld rig doesn’t measure odometry” and “S-DEVO isn’t real-time on high-rate sensors” are testable. As assertions they were correctly rejected. As measurements each becomes a contribution.

| CLAIM WE MADE                    | HOW IT LANDED             | WHAT TO PUBLISH INSTEAD                                                      |
|----------------------------------|---------------------------|------------------------------------------------------------------------------|
| S-DEVO not real-time on desktop  | Factually wrong (R5, R7)  | RTF-vs-event-rate curve on identical hardware; cite their own Orin NX number |
| Small-scale ATE is uninformative | Never stated at all       | Report RPE alongside ATE, plus the sim(3) scale factor $s$ per method        |
| Per-sequence tuning is unfair    | Never stated at all       | Hyperparameters used per method per sequence. Ours: one row                  |
| We are faster                    | No mechanism given (R6.5) | Name each S-DEVO bottleneck and our counterpart, with numbers                |

## 3 Ground truth on S-DEVO

Verified from the RA-L paper (arXiv 2509.08235). **Published in IEEE RA-L, accepted September 2025** — cite the journal, not arXiv (fixes R6.5). Won 1st place in the M3ED SLAM Challenge, 5th CVPR Workshop on Event-based Vision, so beating them on M3ED must be airtight.

### 3.1 Their Table V — computational performance [ms]

| FUNCTION           | DESKTOP             | LAPTOP              | JETSON ORIN NX |
|--------------------|---------------------|---------------------|----------------|
|                    | i7-12700K, RTX 3090 | i5-12500H, RTX 2050 | 25 W           |
| Event voxels       | 13.52               | 40.41               | 121.40         |
| Feature encoder    | 18.39               | 34.13               | 170.51         |
| Patch selector     | 2.92                | 22.28               | 117.60         |
| Depth estimation   | 0.46                | 4.45                | 5.17           |
| Patch graph update | 12.10               | 92.49               | <b>433.64</b>  |
| Bundle adjustment  | 1.53                | 3.03                | 17.40          |
| Others             | 27.33               | 42.84               | 201.51         |
| <b>Subtotal</b>    | <b>62.73</b>        | <b>199.22</b>       | <b>945.83</b>  |

Their own text: “can stably operate in real time at 10 Hz . . . on the desktop platform, and at approximately 5 Hz on the laptop platform. However, the embedded platform’s performance is severely constrained by its 25 W power budget, resulting in significantly higher latency.”

#### The decisive fact

We were directionally right and aimed at the wrong target. “Cannot achieve real-time even on desktop GPUs” is false — 62.73 ms on a 3090 — and that one sentence cost us two reviewers.

But **S-DEVO themselves publish 945.83 ms/frame ( $\approx 1.06$  Hz) on embedded hardware**. Our C++/TRT build runs  $\sim 30$ – $38$  ms on AGX Orin. Even discounting AGX Orin (64 GB, up to 60 W) against Orin NX (25 W) at a generous 2–3 $\times$ , the gap is an order of magnitude — and we do not need to argue, we cite their table and then measure it ourselves on one device.

### 3.2 Their stated bottlenecks map one-to-one onto our fixes

This is the mechanism R6.5 said was missing. Every line is quotable from their paper.

| S-DEVO, THEIR DESCRIPTION                                                    | ORIN NX | DUET-VO COUNTERPART                                             |
|------------------------------------------------------------------------------|---------|-----------------------------------------------------------------|
| Voxelization “entirely on the CPU”                                           | 121.40  | CUDA voxel.cu, GPU-resident                                     |
| “Others ... most of the time spent transferring voxel grids from CPU to GPU” | 201.51  | No H2D voxel transfer; events uploaded raw, voxelized on device |
| Tracking node “primarily implemented in Python”                              | —       | Native C++ with TensorRT fp16 engines                           |
| Patch graph update (dynamic edge lists / scatter)                            | 433.64  | Banded ring buffer, $O(1)$ , torch-scatter-free                 |
| Feature encoder (PyTorch eager)                                              | 170.51  | TensorRT strongly-typed fp16, mtime-cached                      |

This also answers R6.3 verbatim: “Event voxel transfer from CPU to GPU is known to be costly in SDEVO, yet the proposed framework ... reports significantly lower runtime — this discrepancy requires explicit engineering optimization details.”

### 3.3 They report RPE and ATE. We reported only ATE.

Their Table II is [RPE: cm/s, ATE: cm]. We copied the ATE column and dropped RPE. RPE is exactly the metric that exposes a distorted trajectory which ATE flatters — it is our argument about handheld 5 cm rigs, and it was already sitting in the table we were copying from. Restoring it costs nothing and removes the appearance of selective reporting.

### 3.4 Full benchmark inventory — and what we skipped

| DATASET | S-DEVO SEQUENCES                                                                              | WE USED | MISSING           |
|---------|-----------------------------------------------------------------------------------------------|---------|-------------------|
| rpg     | box, monitor, bin, desk, reader                                                               | 0       | all 5             |
| MVSEC   | indoor1/2/3/4 edited                                                                          | 0       | all 4             |
| DSEC    | city04 a–d, city09 a–e, city11 a–b                                                            | 6 (11)  | all 5 city09      |
| VECTor  | robot_normal, corner_slow, hdr_normal, sofa_normal, desk_normal, corridors_dolly, units_dolly | 2 (7)   | all 5 small-scale |
| TUM-VIE | 1d/3d/6d_trans, desk, desk2                                                                   | 0       | all 5             |
| Others  | hnu_campus, drone_fast                                                                        | 0       | both              |

We kept the two hardest VECTor sequences and dropped the five small-scale ones; we kept city04/city11 and dropped all five city09. R5’s “over-representation of large-scale sequences” is literally true of our subset. Fix it by running everything, not by arguing.

## 4 Two papers, split by contribution axis

A parallel RGB system (SDPVO+LC) shares the same backbone, C++/TRT port, circular buffer and loop-closure module. Two papers will come out of this body of work.

### The constraint

The split must run along a **contribution** axis, not a **modality** axis. “Same system, different sensor” is salami-slicing; IEEE RA-L requires disclosure of concurrent related submissions, and the event / patch-based-VO reviewer pool is small enough that both papers may share reviewers. Each paper’s claimed contribution list must be *disjoint*, with explicit cross-citation. Shared implementation is fine when described at different depth for different purposes.

### 4.1 Paper 1 — DUET-VO (this plan)

Stereo **event** VO. Loop closure deliberately excluded: adding it converts a VO paper into a SLAM paper and immediately invites DPV-SLAM / ORB-SLAM3 baselines, loop-detection precision–recall curves and place-recognition ablations — a second full evaluation burden on a paper just rejected for insufficient evaluation. Dropping it is a defensive win, not a sacrifice.

1. **Zero-retraining stereo extension** — dual update operator, shared weights, separate hidden states. (*existing*)
2. **1-D epipolar correlation, reframed.** 1-D correlation-based stereo matching is classical (Lefebvre et al., IVC 2011 — cite it, R7 is right). Our claim is narrower and defensible: (a) it is *necessary* for event features inside a pretrained recurrent update operator, shown by ablation; (b) the zero-padding embedding preserving the operator’s  $7 \times 7 \times P \times P \times 2$  input shape so no retraining is needed; (c) a CUDA kernel with measured speedup.
3. **Horizon-free operation** — banded circular buffer replacing dynamic *ii/jj/kk* edge lists. Constant memory, constant per-frame cost, unbounded sequence length. *Must stay in Paper 1*: it is the direct answer to S-DEVO’s 433.64 ms patch-graph-update line, i.e. the mechanism behind our runtime claim.
4. **Deployable implementation** — native C++, TensorRT fp16, torch-scatter-free, CUDA voxelization and GPU rectification.

#### 4.2 Paper 2 — backbone-native loop closure (RGB and event)

Claim: **loop closure for patch-based VO that reuses the feature maps the backbone already computes** — no BoW vocabulary, no NetVLAD, no extra descriptor network, no additional weights, no additional memory.

Demonstrated on *both* modalities. That is what makes it a general mechanism for patch-based VO backbones rather than “DPVO ported to C++”, and it is what licenses reusing Paper 1’s infrastructure: different claim, cross-cited. It also sidesteps the RGB novelty problem by refusing to compete on RGB accuracy — it does not try to beat MAST3R-SLAM or VGGT-SLAM, it answers a question they structurally cannot: *can loop closure be obtained for free from a VO backbone already running on a 30 W device?*

- Baselines: DBow2/DBow3, NetVLAD, DPV-SLAM’s own loop closure, no-LC ablation.
- Metrics: loop-detection precision/recall, drift reduction at revisit, and the memory and latency cost of the LC module itself.
- Cites Paper 1 for the circular buffer and the stereo extension; re-claims neither.

Ordering: Paper 1 to RA-L as soon as the evaluation is rebuilt; Paper 2 cites it. If the submissions overlap in time, disclose both to the editor.

## 5 Phase 1 — De-risk

2–3 weeks

- ☒ **M1.1** Build S-DEVO on the AGX Orin and confirm it runs. *Do this first.* If it fails, the head-to-head runtime result is gone and the fallback (**M3.4**) becomes load-bearing.
- ☐ **M1.2** Build S-DEVO, DEIO and ESIO/ESVIO on the A5000 box. ESVIO’s ROS1 / Ceres / OpenCV pinning is the usual time sink.
- ☐ **M1.3** Stage datasets: DSEC (11), VECtor (7), TUM-VIE (5), MVSEC (4), rpg (5). **Disk is a constraint** — 337 GB free, DSEC will not fit alongside the rest.
- ☐ **M1.4** Email the baseline authors. Three specific asks, one week, high payoff.
- ☐ Send me the path to the C++ variant with loop closure and the horizon-free buffer — `dpvo2_b/dpvo2/dpvo_cpp` has the ring buffer and TRT engines but no LC module.
- ☐ Re-run the ayasli sequences with the post-fix build (event firing-rate asymmetry). The LOOPCL runs currently on disk are pre-fix.

#### M1.4 — the highest-value email you can send

**DSEC ships no ground-truth trajectories.** S-DEVO derives references from LIO-SAM; so do we. *Two different LIO-SAM runs are two different ground truths* — different parameters, IMU handling, loop closures, origin conventions. An ATE against our reference and an ATE against theirs are not comparable numbers, and no amount of careful running will make them agree.

So ask for: (1) their LIO-SAM reference trajectories; (2) the exact commit and config per dataset; (3) **their raw estimated trajectory files**. Point 3 is the underused one — many authors share `.tum` files they never published. With trajectories you compute ATE, RPE *and* scale yourself, under one protocol, against one reference. Their result, our protocol: strictly better than quoting, and nearly as good as running the code. Worth sending to the S-DEVO, ESVO2 and ES-PTAM authors in the same week.

### 6.1 Provenance discipline

Two *physically separate* tables, so the provenance boundary is a table border rather than a suspicious diagonal through the results:

- **Table A — reproduced by us.** Every cell run by us, protocol fully disclosed, all raw trajectories released. Target: all 34 S-DEVO sequences + 13 M3ED.
- **Table B — literature.** Clearly marked and cited, with an explicit note that accumulation windows and alignment conventions may differ.

Never a bare dash. Every empty cell carries a reason code: `div` (ran, diverged) · `nc` (no public code) · `ng` (no ground truth) · `oom` (resource limit) · `ns` (sensor modality absent). State the fraction in the caption: “*N of M cells (X%) produced by us; all raw trajectories released at ...*”

Columns per method: **RPE (cm/s)**, **ATE (cm)**, **sim(3) scale *s***.

#### The scale column is the quiet weapon

A method requiring  $s = 1.4$  is not metric. The  $s$  column turns “their sim(3) alignment is unfair” from a complaint into a number sitting in our own table, where the reader draws the conclusion themselves. It also cannot be computed from a published table cell — only from a trajectory. Which is the real argument for running baselines rather than quoting them: **quoted numbers are inert, you cannot ask them questions.**

### 6.2 Small-scale sequences: include all of them

The original instinct was to omit `rpg` / `MVSEC` / `small VECtor` / `TUM-VIE` because a handheld rig moving in a few-centimetre volume produces a flattering ATE regardless of trajectory shape. **The observation is correct; using it as grounds for omission is not.** We win on those sequences on both ATE and RPE, so including them costs nothing and removes the single most damaging objection in the review.

Keep the methodological point, restated as a caveat on interpretation rather than an excuse for omission — and now backed by measurement:

*On sequences below ~10 m of travel, inter-method ATE differences fall within intra-method run-to-run variance; ATE alone therefore does not discriminate, and we report RPE and sim(3) scale alongside it.*

Evidence already exists: three runs of an identical configuration on `ayasli_loop_1` gave ATE 226 / 377 / 625 cm — a 2.8× spread. Made as a reason to omit data this is special pleading; made as a reason to add metrics it is rigour. Where a win on a short sequence falls inside the interval, say so.

### 6.3 Baseline protocol — be conspicuously generous

#### One rule

Every baseline runs at its **author-recommended default** — that is what makes their published numbers comparable in the first place. We never re-tune a baseline downward, and we never raise our own patch count to match theirs. **DUET-VO reports exactly two fixed configurations** (default 96, superfast 24), held constant across every sequence in the paper.

**S-DEVO on M3ED.** Their paper reports no M3ED results at all, while their conclusion claims 1st place in the M3ED SLAM Challenge — so M3ED numbers exist somewhere but were never published, presumably with challenge-specific tuning. State exactly that, neutrally, with both citations. Their released configuration does not transfer: it produces scale failures, and recovery needs per-sequence tuning of `SEARCH_RANGE`, with different values indoors and outdoors. Handle it by being maximally generous.

- ☐ **M2.1** DUET-VO on all 47 sequences, 10 runs, 2 fixed configs.
- ☐ **M2.2** S-DEVO, DEIO, ESIO/ESVIO on the same 47, author defaults.
- ☐ **M2.3** Sweep `SEARCH_RANGE` per M3ED sequence; report S-DEVO’s *per-sequence best*. Give them their oracle — if DUET-VO beats an oracle-tuned S-DEVO the result is unimpeachable, and it answers R5’s tuning objection by having done that tuning exhaustively.
- ☐ **M2.4** Generate Table A and Table B, with reason codes and median [min, max].

- ☐ **M2.5** Discrimination-threshold figure: inter-method ATE gap vs. intra-method run-to-run spread, against trajectory length, crossover marked.
- ☐ Figure: ATE vs. SEARCH\_RANGE, indoor vs. outdoor, showing the optimum shifts. This is the per-sequence-tuning argument as a measurement rather than a complaint.
- ☐ Acknowledge the S-DEVO authors for parameter guidance. **Do not quote the private correspondence** — ethically fraught without permission, reads as adversarial, and unnecessary given the public record.
- ☐ Hyperparameter table, per method per sequence. Ours is one row; that is the argument, made visually.
- ☐ ESVIO on DSEC city09 night sequences — it is *image-aided*, not event-only, so characterise the RGB-degradation failure directly rather than asserting it.

## 7 Phase 3 — Runtime evidence

1–2 weeks

### 7.1 Hardware mapping onto S-DEVO's three platforms

We do not own their exact parts. State the substitution explicitly and quantify it — a reviewer who spots an unstated mismatch treats it as another provenance failure.

| TIER     | THEIRS                                 | OURS                                    | WHAT TO PRINT                                                                                            |
|----------|----------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------|
| Desktop  | RTX 3090 (GA102, 10496 cores, 35.6 TF) | <b>RTX A5000</b> (GA102, 8192, 27.8 TF) | <i>Same GA102 die</i> , one SM tier down — $\sim 78\%$ of the 3090's fp32 throughput. Report the ratio.  |
| Ceiling  | —                                      | <b>RTX 6000 Ada</b> (48 GB, 91 TF)      | Shows the trend is not architecture-specific.                                                            |
| Laptop   | RTX 2050 (Turing, 2048)                | <b>RTX 3070 Laptop</b> (Ampere, 5120)   | Ours is $\sim 2.5\text{--}3\times$ <i>faster</i> , so any S-DEVO slowness measured here is conservative. |
| Embedded | Orin NX 16 GB, 25 W, 1024 cores        | <b>AGX Orin 64 GB</b> , 2048 cores      | Not the same part — ours is stronger. Handled below.                                                     |

#### The Orin NX gap, and why it stops mattering

Comparing our AGX Orin against their Orin NX would be exactly the kind of unstated mismatch that got us rejected. Two moves remove the problem:

- Run S-DEVO on our own AGX Orin.** Both systems then sit on an identical device and their published Orin NX figure becomes supporting context rather than the basis of the comparison. *This is the primary result* — and **M1.1**.
- Power-capped sweep as the bridge.** `nvpmodel` exposes 15/30/50 W/MAXN and core capping. A  $\sim 25$  W, 1024-core envelope gives a point directly comparable to their 25 W row — and engages their stated excuse (“severely constrained by its 25 W power budget”) with data rather than rhetoric.

- ☐ **M3.1** Mirror S-DEVO's Table V *row-for-row*, same function names, on all four platforms. Add the rows they omit and reviewers demanded: rectification/undistortion (R5), host→device transfer as its own line, feature extraction/patchify (R6.1).
- ☒ **M3.2** S-DEVO measured on our AGX Orin — the primary head-to-head. Depends on **M1.1**.
- ☐ **M3.3 RTF vs. event-rate figure** — the figure that makes the paper.  $x$ : Mev/s.  $y$ : ms/frame, with the real-time line. Both methods, across DAVIS346 260p → DSEC 480p → M3ED 720p → Prophesee. Plot their published operating point as a labelled marker. Nobody has characterised event-VO cost against event rate; this is publishable on its own and impossible to rebut.
- ☐ **M3.4** Power sweep on AGX Orin: ATE and RTF at 15/30/50 W/MAXN, plus the 25 W core-capped bridge point.
- ☐ **M3.5** Tail latency: p50 / p95 / p99 and worst case. A real-time claim backed only by a mean is not a real-time claim.

## 8 Phase 4 — Ablations

1 week



- ☐ **M4.1** Patch-count Pareto sweep 12/24/32/48/64/96/192 → ATE vs. latency frontier, with baselines overlaid as scattered points. R7 asked for exactly this instead of two isolated configurations.
- ☐ **M4.2** 1-D vs. 2-D correlation *at each* patch count.
- ☐ **M4.3** Horizon-free ring buffer vs. full sparse graph — memory and per-frame cost against sequence length.
- ☐ **M4.4** TRT fp16 vs. fp32 vs. PyTorch eager. Prove the speed did not come out of accuracy, before a reviewer asks.
- ☐ Disparity shortcut vs. full SE(3) reprojection (*carried over*).

## 9 Phase 5 — Manuscript

3–4 weeks

Cheapest points in the entire plan. Roughly one week of work answers a third of the review.

### 9.1 M5.1 Factual — non-negotiable

- ☐ **Delete** “cannot achieve real-time even on desktop GPUs.” False, and it cost us the paper.
- ☐ Replace with their own number: 945.83 ms on Orin NX, cited to their Table V, plus our measurement on identical hardware.
- ☐ Add the mechanism paragraph (§3.2) — their CPU-only voxelization, CPU→GPU transfer, Python tracking node, 433 ms patch-graph update, each against our counterpart. This is R6.5.
- ☐ Fix “sensor-rich fusion . . . at the cost of reduced accuracy” — contradicts ESIVO’s own results (R7.a).
- ☐ §II.A describes EMVS but cites EVO. Cite Rebecq et al., IJCV 2018 (R7.b).

### 9.2 M5.2 Structure, citations, terminology

- ☐ Drop the standalone event-representation subsection; merge II.D into II.B; expand model-based event odometry coverage; compress III.A–C to overview + inherited + new (R6.1, R6.2).
- ☐ Cite published venues, not arXiv — **S-DEVO is IEEE RA-L 2025**; same for DEVO, DEIO (R6.5).
- ☐ Add Lefebvre et al., IVC 29(9), 2011 and reframe the 1-D claim. Add Ghosh & Gallego, AIS 4(12), 2022 (R7).
- ☐ Expand GRU on first use; unify “GRU” and “Dual Update” across text, Fig. 2 and Table IV (R6.6, R6.4.2).
- ☐ Define “sliding window” and “superfast” at first use. Replace “profound performance” (R7).

### 9.3 M5.3–M5.6 Integration and release

- ☐ **M5.3** Integrate all new tables and figures.
- ☐ **M5.4 Code release tag with the LC module stripped, not merely disabled.** If reviewers find an unused LC module in released code while we do not claim LC, they will ask why. Restore for Paper 2. State the release in the abstract.
- ☐ **M5.5** Co-author review round.
- ☐ **M5.6** Submit.

#### If a reproduction disagrees with a published number

First diagnose: is the gap *uniform* across sequences (your pipeline differs systematically — suspect GT source, alignment convention, trimming, or which trajectory file is scored) or *sequence-specific* (your harness is validated and the divergent ones are credible)? Use the sequences you *can* reproduce to certify the harness — that is what makes the ones you cannot believable.

Then report both: **published value in the main table** (their best claim, so nobody can say we handi-capped a competitor), **our measurement in a separate reproducibility table** with full protocol. Never build a load-bearing claim on a reproduction being worse than a published number — that is structurally the same mistake as the real-time sentence. Never speculate in print about *why* their numbers are better; show the sweep and let the data say it.

## 10 Open items

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- ☐ Ground truth for the event sequences (ayasli\_loop\_\*) — RTK or LiDAR? The RTK pipeline is documented for the ZED/RGB runs only. **Note:** single-antenna RTK carries no attitude (identity quaternions), so no rotation metric is valid against it — the harness suppresses that column rather than emit a plausible-looking number.
  - ☐ Clock skew:  $\sim 0.8$  s measured on the event rig,  $\sim 1.2$  s documented for ZED. Whatever offset is applied must be *reported*, never silent.
  - ☐ Disk: 337 GB free against DSEC + VECtor + TUM-VIE + MVSEC + rpg. Stage, or add storage.

#### Already built — do not redo

- ☒ paper\_related/RESUBMISSION\_PLAN.md — the source of this document.
- ☒ eval/benchmark/metrics.py — canonical ATE + time-based RPE + sim(3) scale + median/[min,max], validated against the RTK data. Implements RPE over wall-clock deltas because evo supports only frame/metre/angle deltas and event keyframe rates are non-uniform.
- ☒ eval/benchmark/literature/sdevo\_ra12025.json — all 34 Table II sequences, Table III, Table V, machine-readable, with their two internal inconsistencies recorded (city09\_d ATE and city09\_e RPE disagree between their own tables).

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Decided: RA-L rolling · Paper 1 excludes loop closure · full public code release · reproducible baselines DUET-VO / S-DEVO / DEIO / ESIO-ESVIO (Table A); ESVO / ES-PTAM / ESVO2 from literature (Table B).